AWS VPC PEERING

Architecture, Configuration & Validation Report

EWB Cloud Labs | AWS Region: ap-south-1 (Mumbai)
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Report date: 30 October 2026

PROJECT SCOPE

Private connectivity between two Amazon VPCs using VPC Peering, route tables, security-group controls, and bastion-based SSH access.

1. Executive Summary

This report documents an AWS VPC peering implementation in the Mumbai region (ap-south-1). The design connects two VPCs with non-overlapping IPv4 CIDR ranges and uses explicit routes to enable private communication between application instances. A bastion host provides an administrative entry point to private instances. Security groups restrict SSH and ICMP traffic to designated source addresses.

Validation evidence recorded for the project shows successful ICMP connectivity in both directions when the intended secondary private IP addresses are explicitly selected as source addresses. Each test reports four packets transmitted and received, with 0% packet loss. Items for which screenshots or additional evidence were not supplied are marked for completion.

2. Objectives and Scope

- Create two VPCs with non-overlapping address ranges.
- Create and associate public/private subnets and configure the required Internet Gateway.
- Establish VPC peering and configure reciprocal routes.
- Restrict instance-level traffic with security groups.
- Access private instances through a bastion host using SSH ProxyJump.
- Validate private connectivity in both directions and document evidence.
- Record cleanup steps and evidence still required for a complete submission.

3. Architecture Overview

VPC A (10.10.0.0/16) contains a public subnet for the bastion and a private subnet for App-A. VPC B (10.20.0.0/16) contains a private subnet for App-B. The VPC peering connection carries private traffic between the two CIDR ranges; it does not provide transitive routing.

Component	Address / configuration	Purpose
VPC A	10.10.0.0/16	Bastion and App-A network
VPC A public subnet	10.10.1.0/24	Bastion entry point
VPC A private subnet	10.10.2.0/24	App-A workload
VPC B	10.20.0.0/16	App-B network
VPC B private subnet	10.20.1.0/24	App-B workload
VPC peering	pcx-01934796beac9a6c4	Private VPC-to-VPC routing
AWS Region	ap-south-1	Mumbai

Conceptual traffic path

```
Administrator / Kali → Bastion (public subnet, VPC A)
                         ├── SSH → App-A (10.10.2.10)
                         └── SSH → App-B (10.20.1.10 via peering)

App-A (VPC A private subnet) ⇄ VPC Peering ⇄ App-B (VPC B private subnet)
```

The path is conceptual. Verify current public IPs, subnet associations, and instance addresses in the AWS console before final submission.

4. Resource Inventory

Resource	Name / ID	Notes
VPC A	ewb-vpc-a / vpc-08092d12e8b63a296	CIDR 10.10.0.0/16
VPC B	ewb-vpc-b / vpc-0cabf8113fbdd7d46	CIDR 10.20.0.0/16
Peering connection	ewb-a-b-peer / pcx-01934796beac9a6c4	Confirm status is Active

Internet Gateway	ewb-igw-a / igw-030431e12118d6dad	Attached to VPC A
Public route table	ewb-rt-a-public / rtb-01b8e85216ac7c4d2	IGW default route and peering route
Private route table A	ewb-rt-a-private / rtb-07be1132cea4a051d	Associated with App-A subnet
Private route table B	ewb-rt-b-private / rtb-0e06e4260a368e14d	Associated with App-B subnet
App-A instance	i-0c275e8611a22903a	Primary 10.10.2.53; secondary intended 10.10.2.10
App-A ENI	eni-0a7048efe76a299c5	Subnet subnet-0558eab73a3d67ba1
App-B instance	i-02eb907d7b30eaac6	Primary 10.20.1.97; secondary intended 10.20.1.10
App-B ENI	eni-0427654500ae809fc	Subnet subnet-0b1529b9710d77133
App-B security group	sg-09d1174a8b11ca1d9	Name: ewb-sg-app-b
Bastion	Instance ID not recorded	Private IP 10.10.1.26; reported public IP 13.207.38.241
App-A security group	ID not recorded	Name: ewb-sg-app-a
Bastion security group	ID not recorded	Name: ewb-sg-bastion

Inventory note: Some IDs and IP addresses were not available in the supplied notes. Replace missing values and re-check dynamic public IPs against the current AWS console.

5. Routing and Peering Configuration

5.1 Required route entries

Route table	Destination	Target
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	10.20.0.0/16	pcx-01934796beac9a6c4
ewb-rt-a-public	0.0.0.0/0	igw-030431e12118d6dad
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	pcx-01934796beac9a6c4
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	pcx-01934796beac9a6c4

5.2 Configuration checks

- The peering connection must be Active.
- Both VPC CIDR blocks must be non-overlapping.
- The route table associated with each workload subnet must include the reciprocal peer CIDR route.
- The public subnet route table must have a default route to the Internet Gateway if the bastion requires direct public access.
- VPC peering is not transitive; a peered VPC does not automatically provide onward access to a third network.

6. Security Controls

Security group	Intended inbound rule	Purpose
ewb-sg-bastion	SSH TCP/22 from administrator's current public IPv4 /32	Restrict administrative entry
ewb-sg-app-a	SSH TCP/22 from 10.10.1.26/32	Allow SSH from bastion
ewb-sg-app-a	ICMP Echo Request from 10.20.1.10/32	Allow App-B ping to App-A
ewb-sg-app-b	SSH TCP/22 from 10.10.1.26/32	Allow SSH from bastion
ewb-sg-app-b	ICMP Echo Request from 10.10.2.10/32	Allow App-A ping to App-B

Security groups are stateful. The ICMP entries are intended to permit Echo Requests from the opposite workload IP. Confirm outbound rules and any additional required traffic in the live configuration. Avoid exposing SSH to 0.0.0.0/0.

7. Administrative Access Through Bastion

The reported access method uses SSH ProxyJump, allowing an administrator to connect to private instances through the bastion without assigning public IP addresses to the application instances.

```
Host ewb-bastion
    HostName 13.207.38.241
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes

Host ewb-app-a
    HostName 10.10.2.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion

Host ewb-app-b
    HostName 10.20.1.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

Example commands:

```
ssh ewb-bastion
ssh ewb-app-a
ssh ewb-app-b
```

Verify the bastion's current public IP, key-file permissions, OS login username, and actual instance IPs before using this example configuration.

8. Connectivity Validation

The following test commands and results were recorded for the private secondary IP addresses. The explicit source address matters because the instances also have primary private IPs.

8.1 App-A to App-B

```
ping -I 10.10.2.10 -c 4 -W 2 10.20.1.10
```

Metric	Recorded result
Packets transmitted	4
Packets received	4
Packet loss	0%
Average RTT	0.177 ms

8.2 App-B to App-A

```
ping -I 10.20.1.10 -c 4 -W 2 10.10.2.10
```

Metric	Recorded result
Packets transmitted	4
Packets received	4
Packet loss	0%
Average RTT	0.167 ms

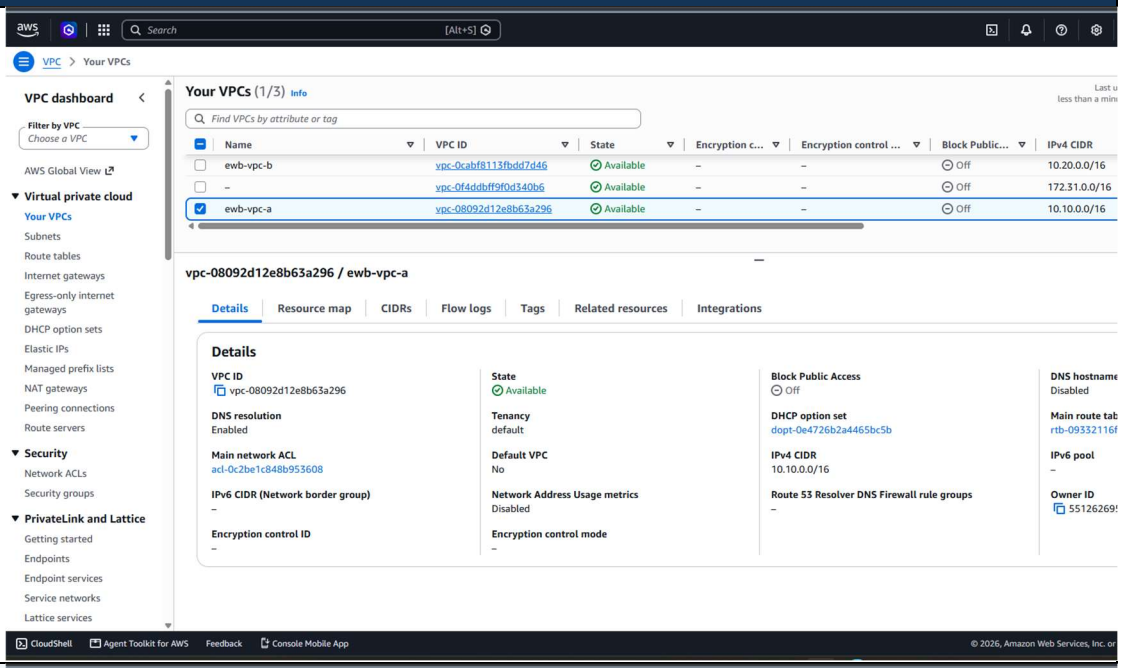
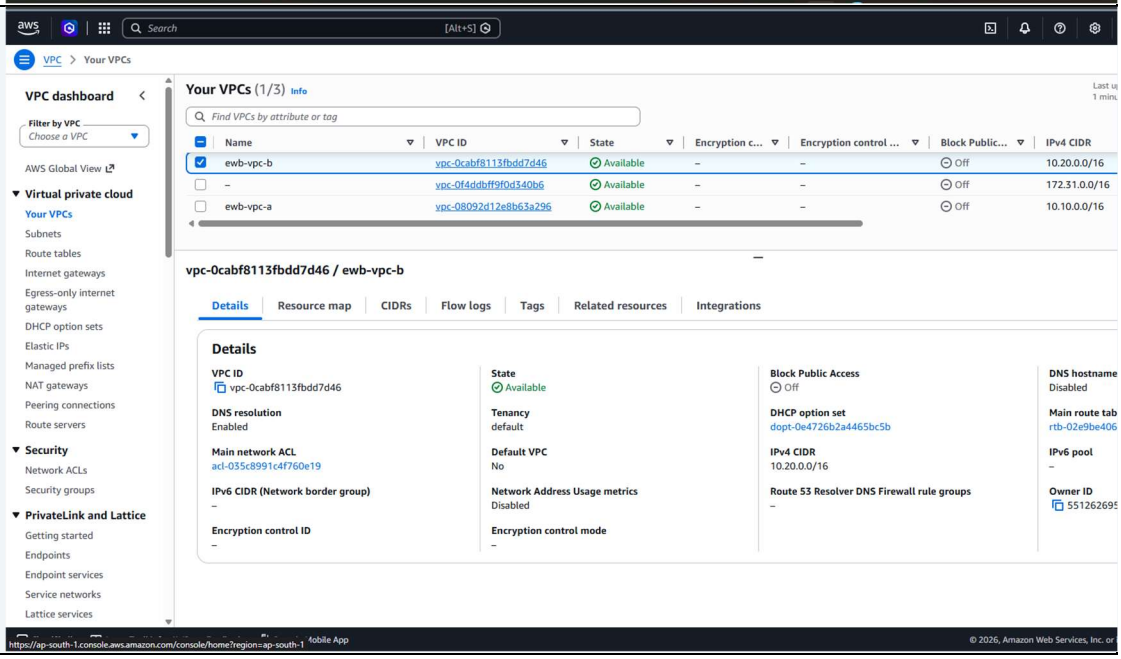
8.3 Interpretation and limitations

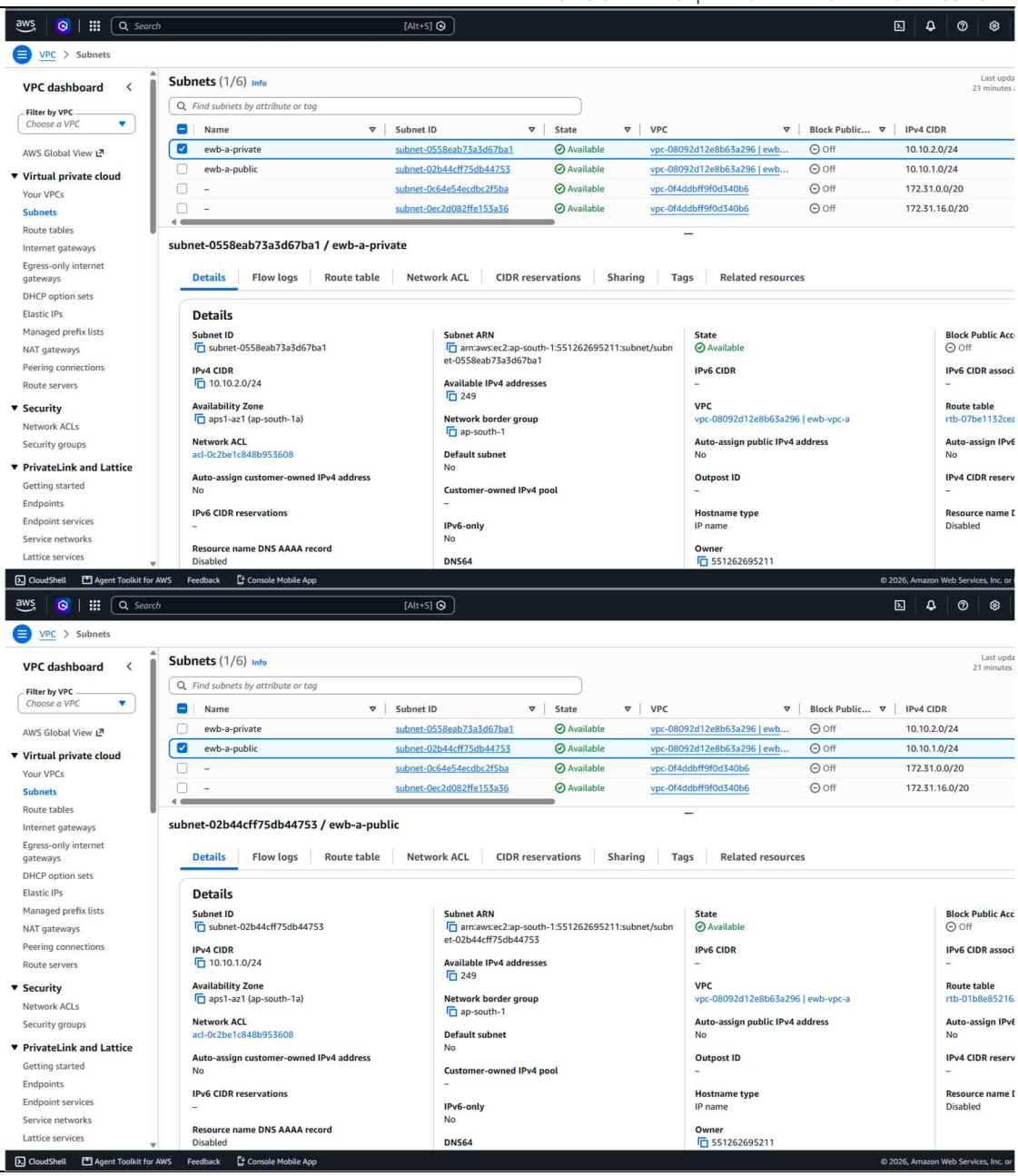
The recorded results demonstrate bidirectional ICMP reachability using the specified secondary private IP addresses. They do not by themselves prove that all security-group rules, SSH paths, failure controls, or cleanup steps have been captured in screenshot evidence.

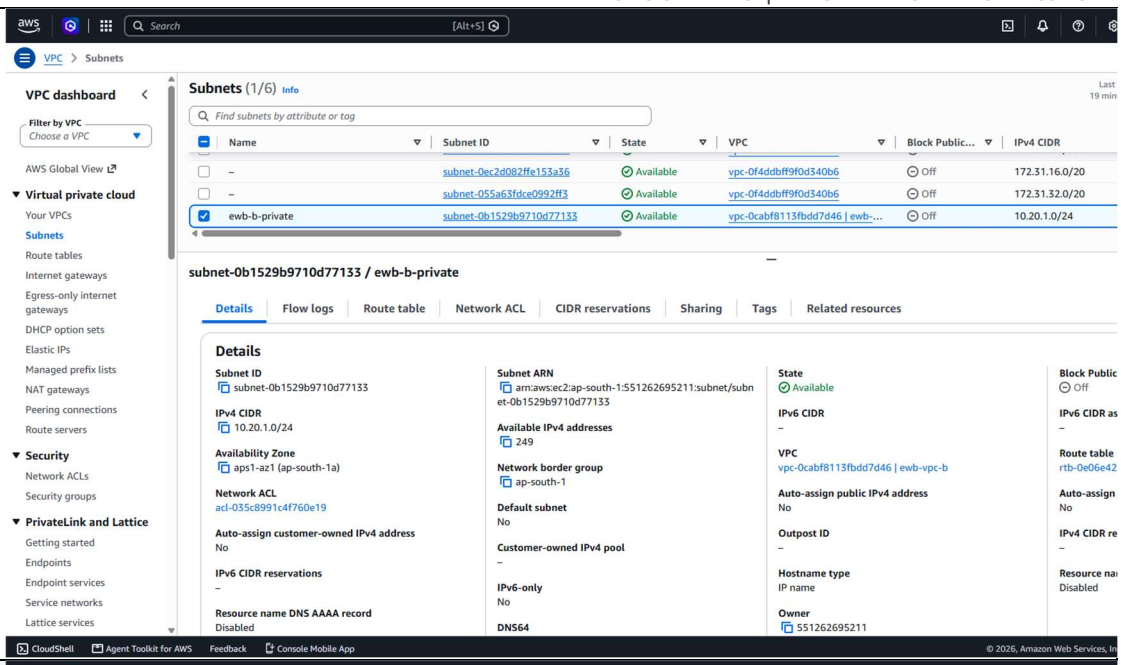
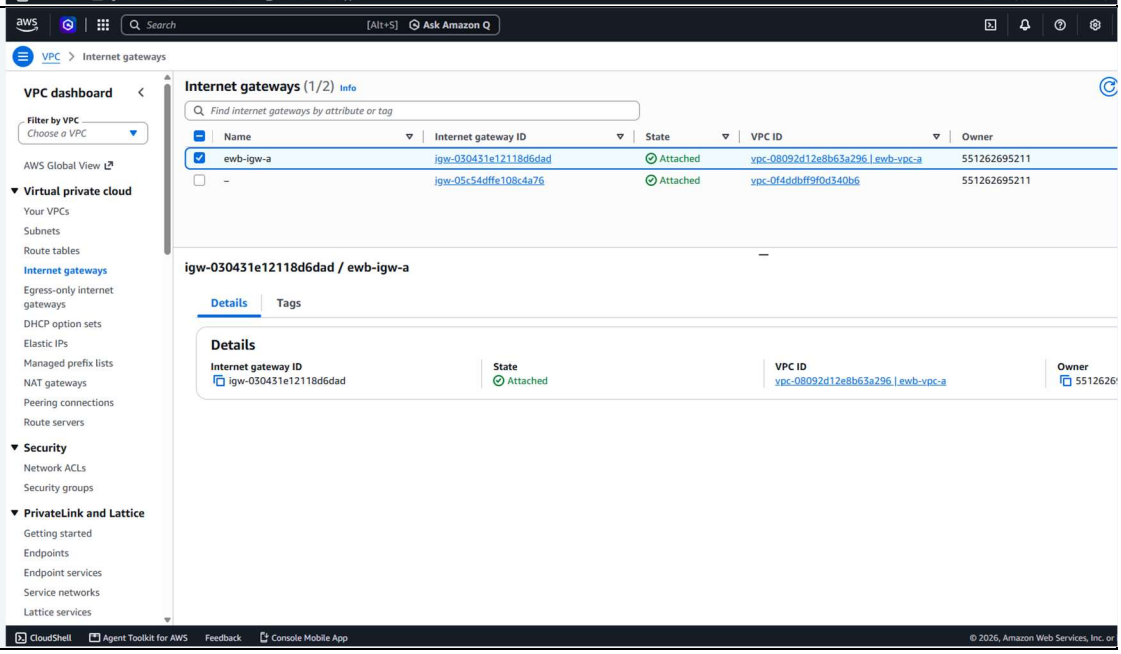
The primary private addresses recorded for App-A (10.10.2.53) and App-B (10.20.1.97) differ from the intended secondary addresses used in the ping tests. The narrow ICMP rules must match the actual source address used by the test.

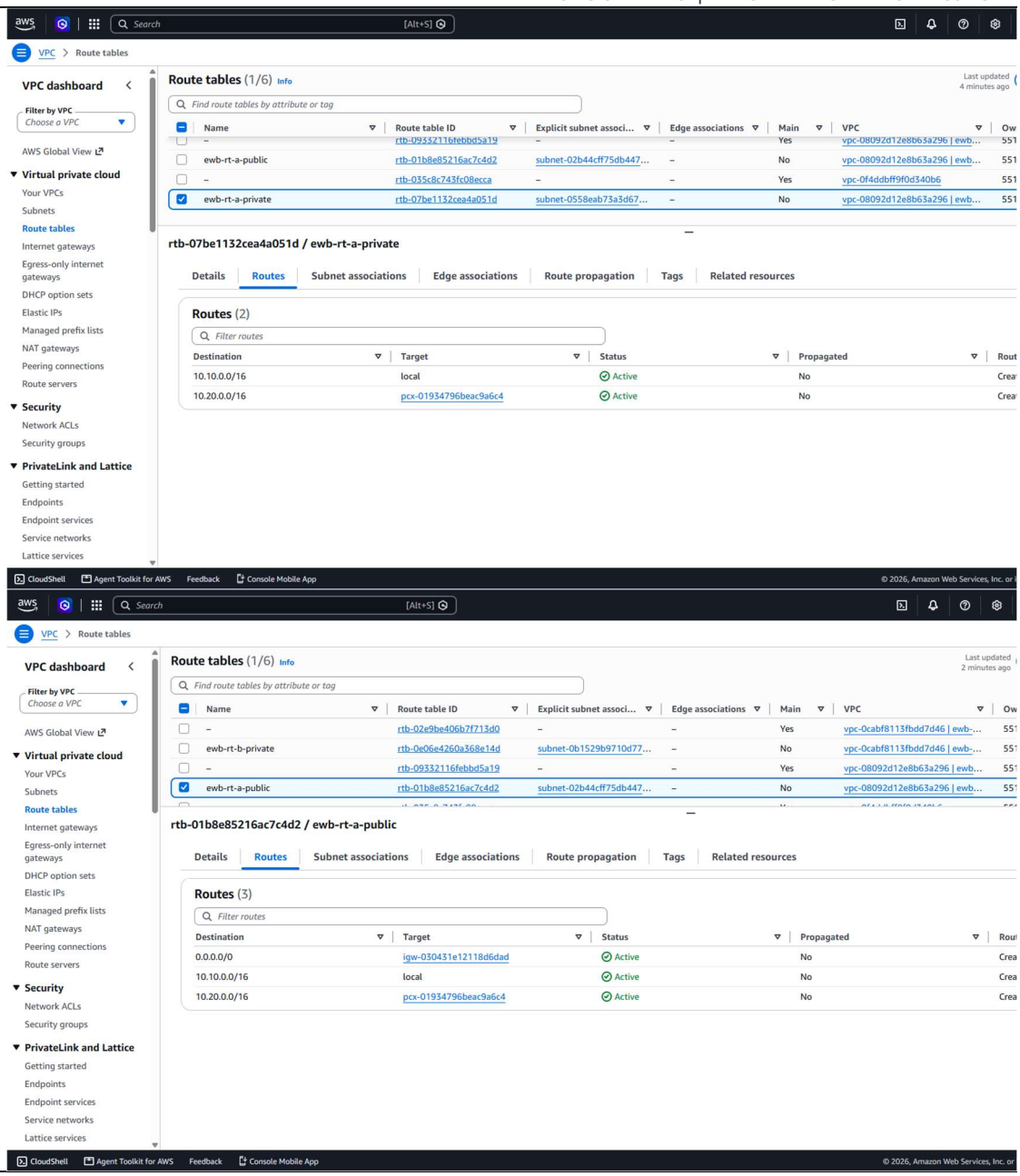
9. Screenshot Evidence Register

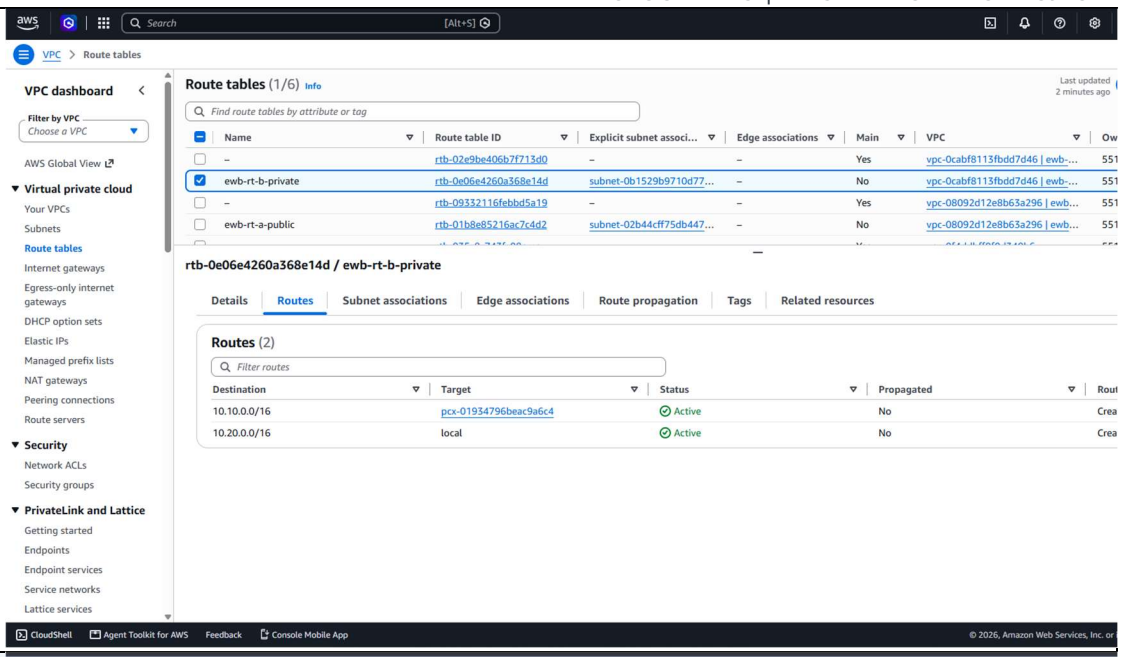
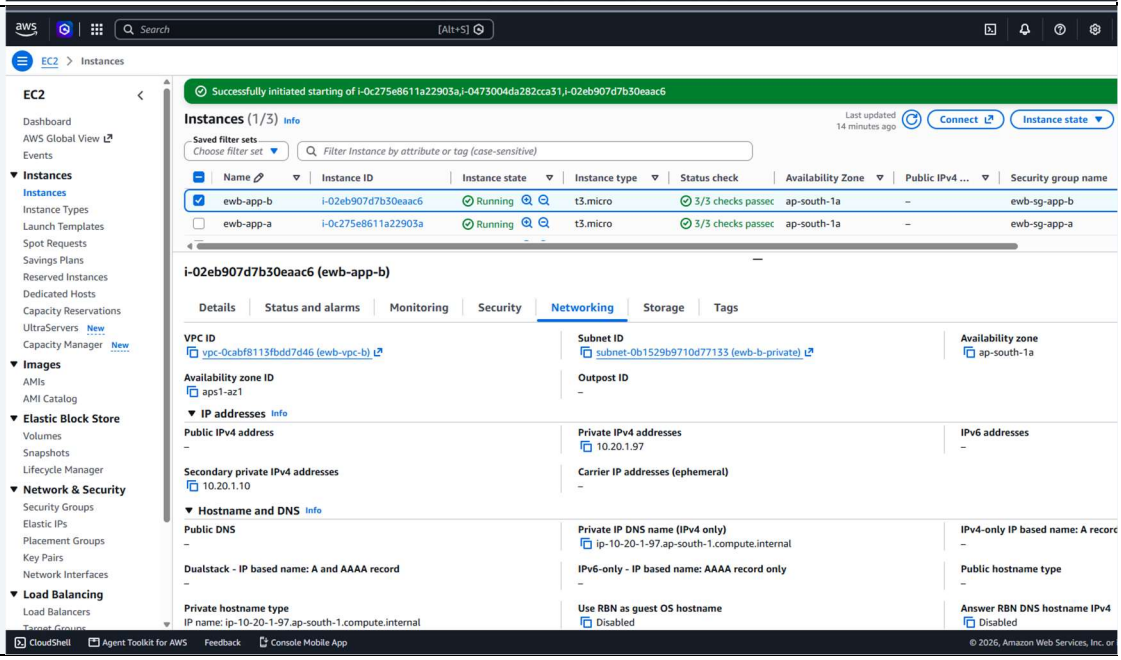
Add genuine screenshots from the AWS console and terminal. The screenshots were not supplied as image files for this editable report, so no evidence image is fabricated or embedded here.

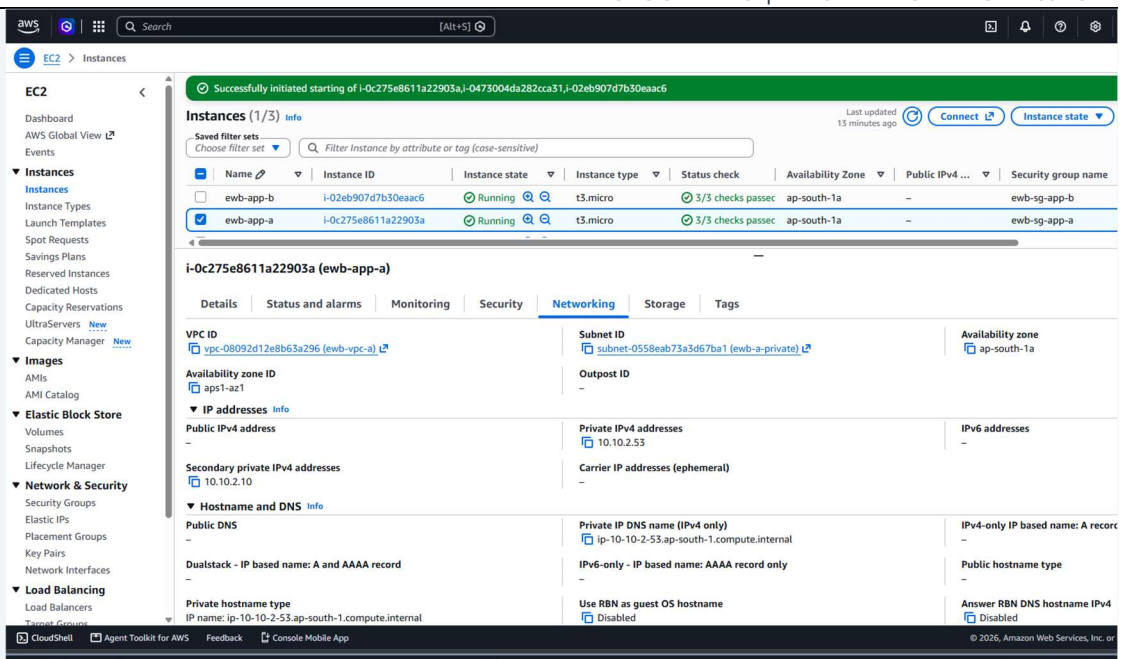
Filename	Evidence to capture	Status
01-vpc-a-created.png	VPC A name, CIDR, and ID	
02-vpc-b-created.png	VPC B name, CIDR, and ID	

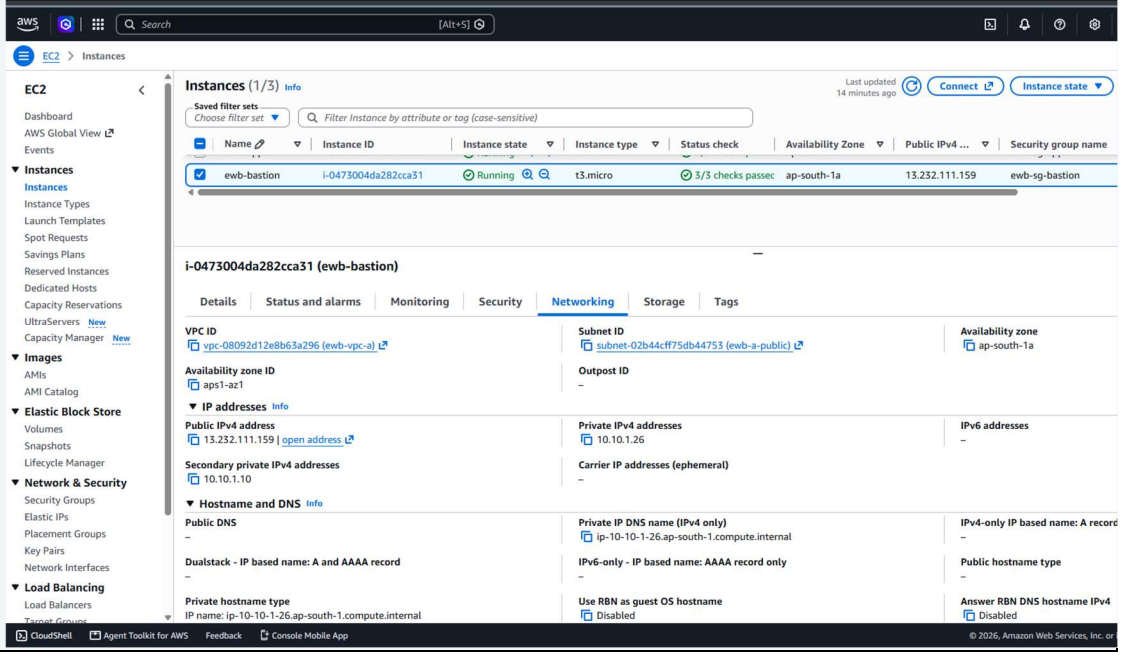
03-subnets-created.png	Subnet CIDRs and VPC associations	

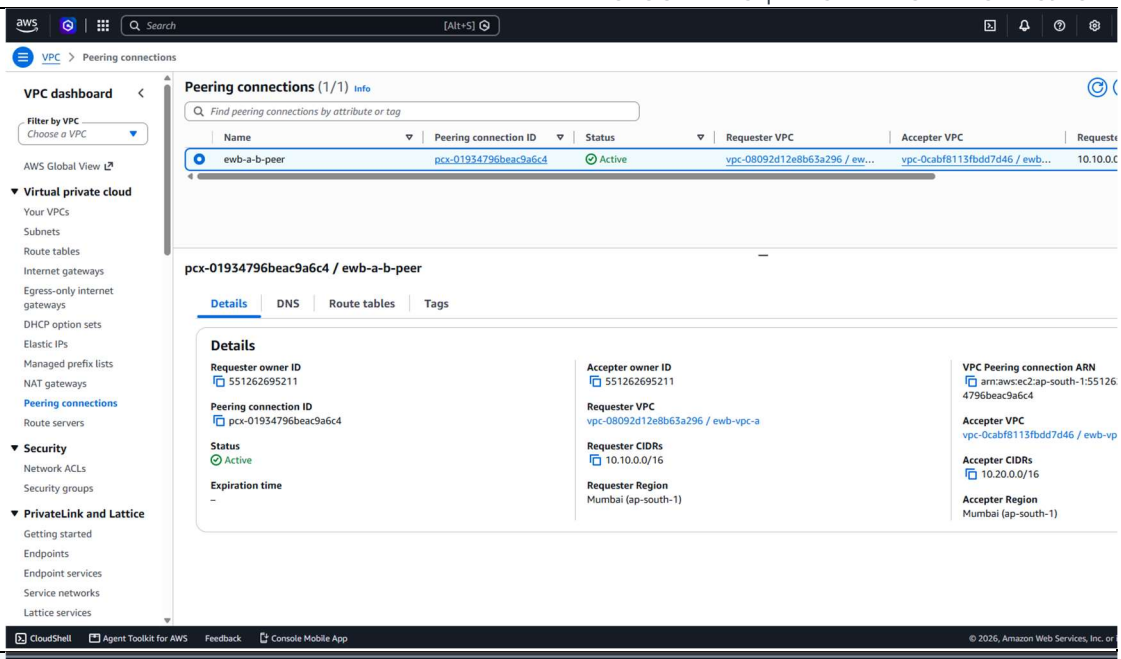
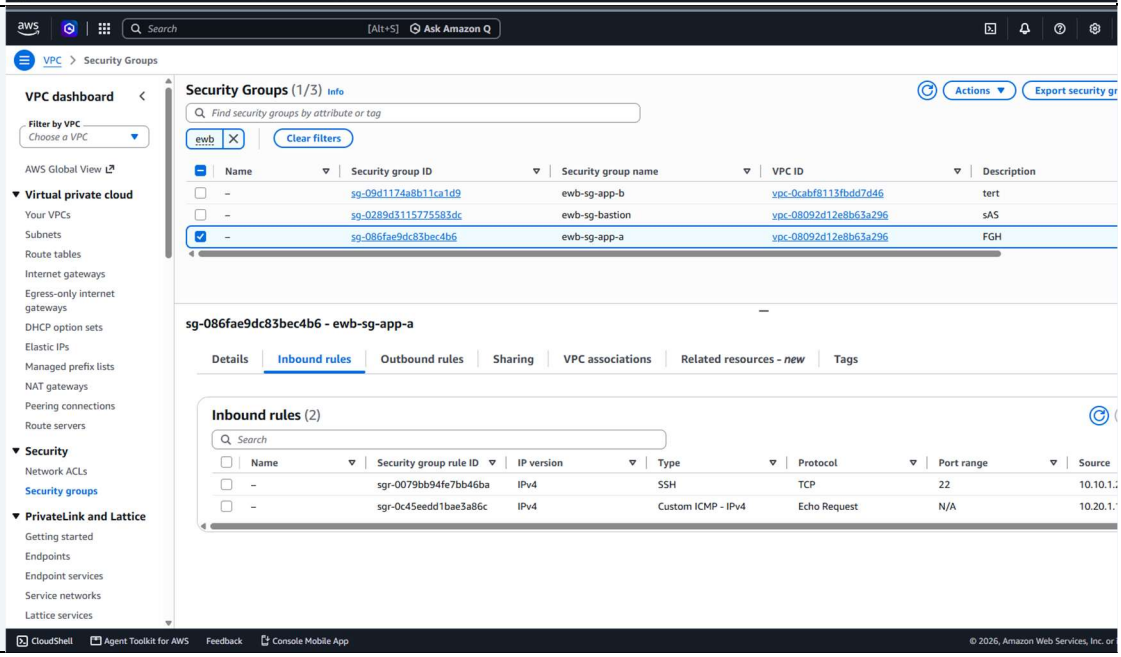
		
04-internet-gateway.png	IGW attached to VPC A	

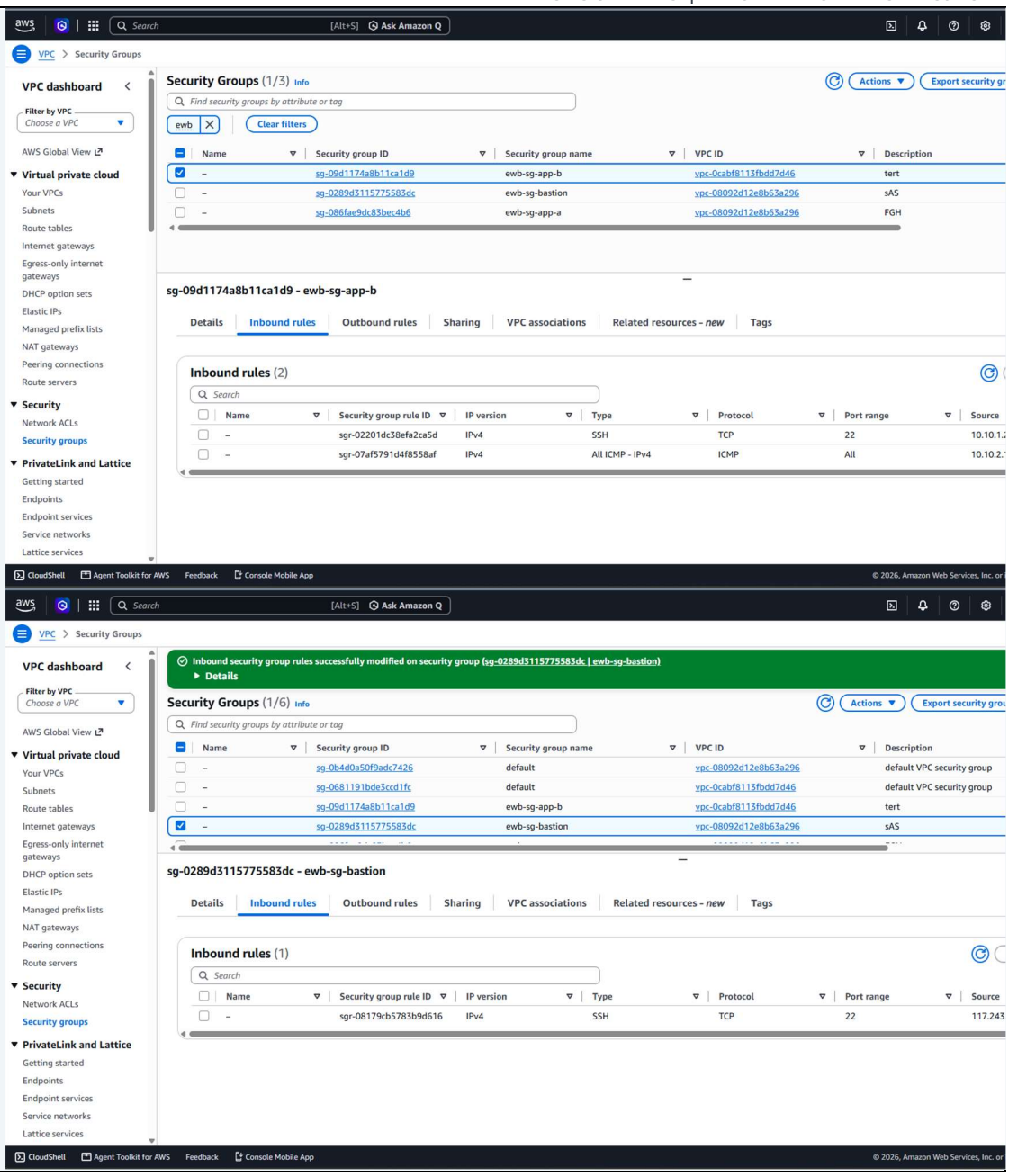
05-route-tables.png	Routes and subnet associations	

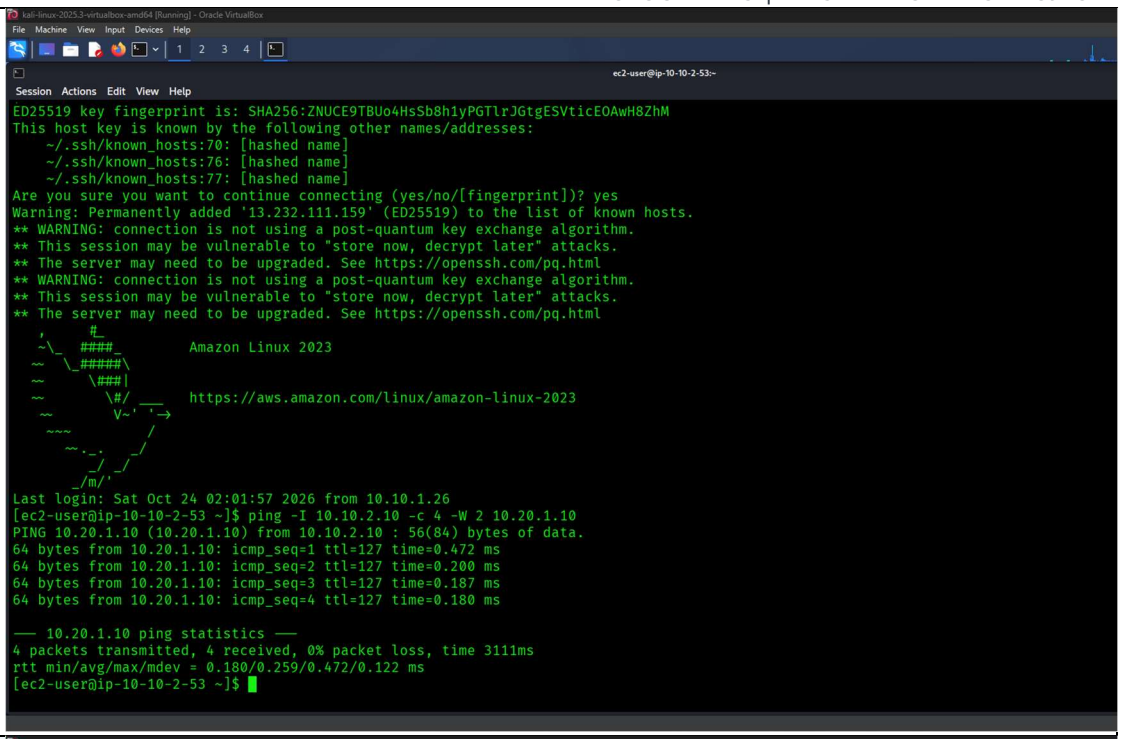
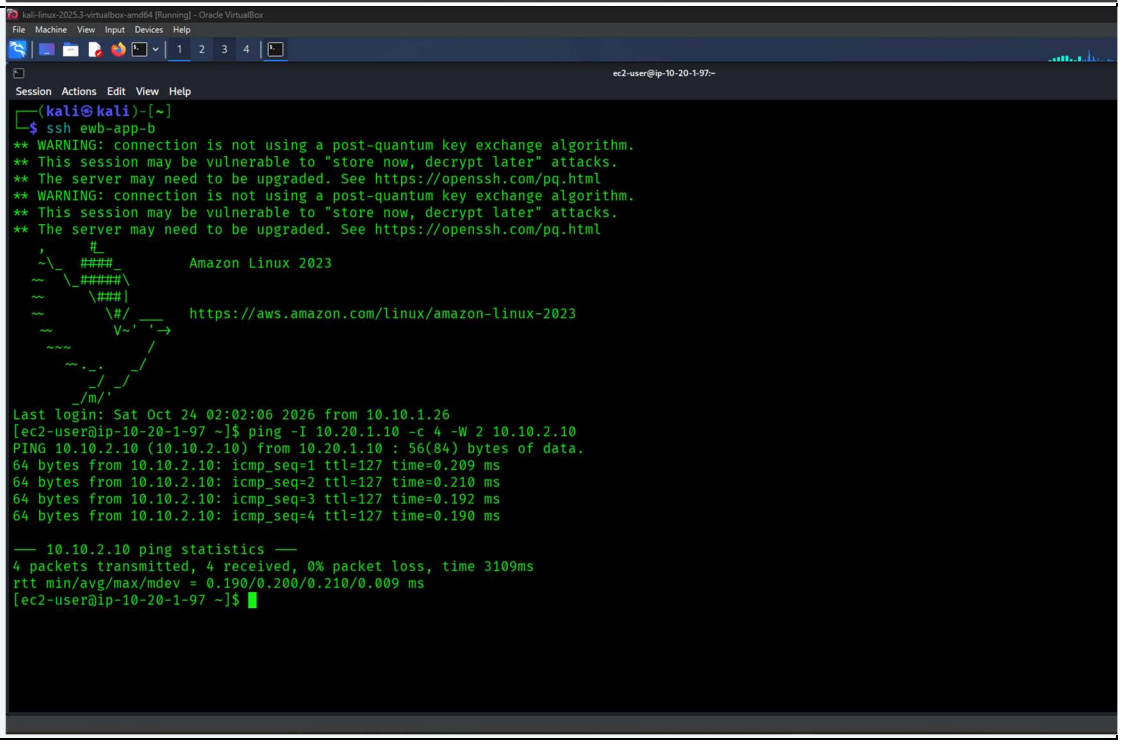
		
06-ec2-instances.png	Instance IDs and private addresses	

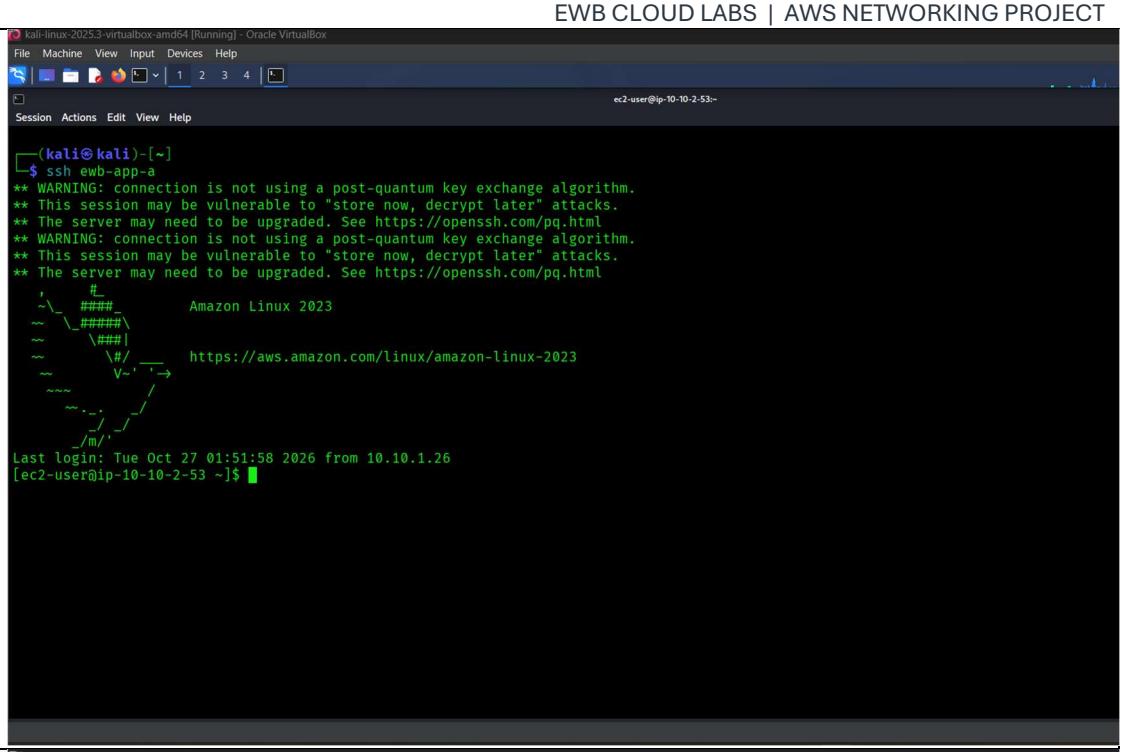
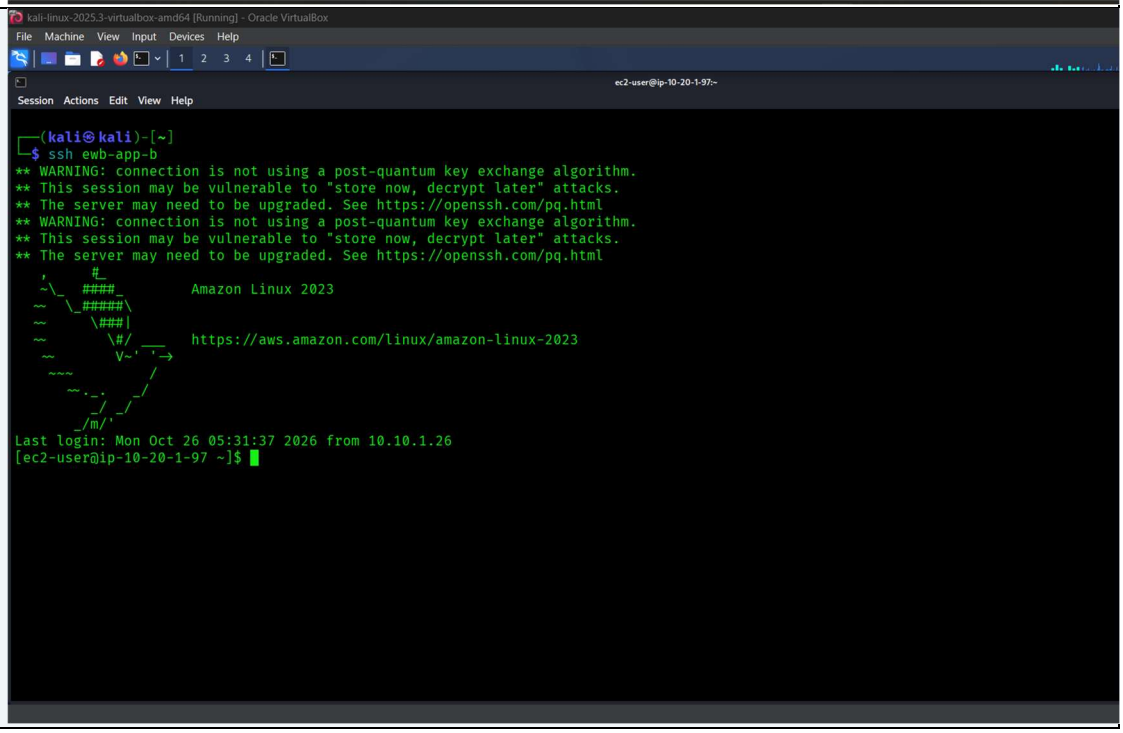

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Capacity Manager New
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Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Successfully initiated starting of i-0c275e8611a22903a,i-0473004da282cca31,i-02eb907d7b30eaac6
Instances (1/3) Info
Last updated 13 minutes ago
Connect
Instance state
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 ...
Security group name
ewb-app-b
i-02eb907d7b30eaac6
Running
t3.micro
3/3 checks passed
ap-south-1a
–
ewb-sg-app-b
ewb-app-a
i-0c275e8611a22903a
Running
t3.micro
3/3 checks passed
ap-south-1a
–
ewb-sg-app-a
i-0c275e8611a22903a (ewb-app-a)
Details
Status and alarms
Monitoring
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Tags
VPC ID
vpc-08092d12e8b63a296 (ewb-vpc-a)
Subnet ID
subnet-0558eab73a3d67ba1 (ewb-a-private)
Availability zone
ap-south-1a
Availability zone ID
aps1-az1
Outpost ID
–
IP addresses Info
Public IPv4 address
–
Private IPv4 addresses
10.10.2.53
IPv6 addresses
–
Secondary private IPv4 addresses
10.10.2.10
Carrier IP addresses (ephemeral)
–
Hostname and DNS Info
Public DNS
–
Private IP DNS name (IPv4 only)
ip-10-10-2-53.ap-south-1.compute.internal
IPv4-only IP based name: A record
–
Dualstack - IP based name: A and AAAA record
–
IPv6-only - IP based name: AAAA record only
–
Public hostname type
–
Private hostname type
IP name: ip-10-10-2-53.ap-south-1.compute.internal
Use RBN as guest OS hostname
Disabled
Answer RBN DNS hostname IPv4
Disabled
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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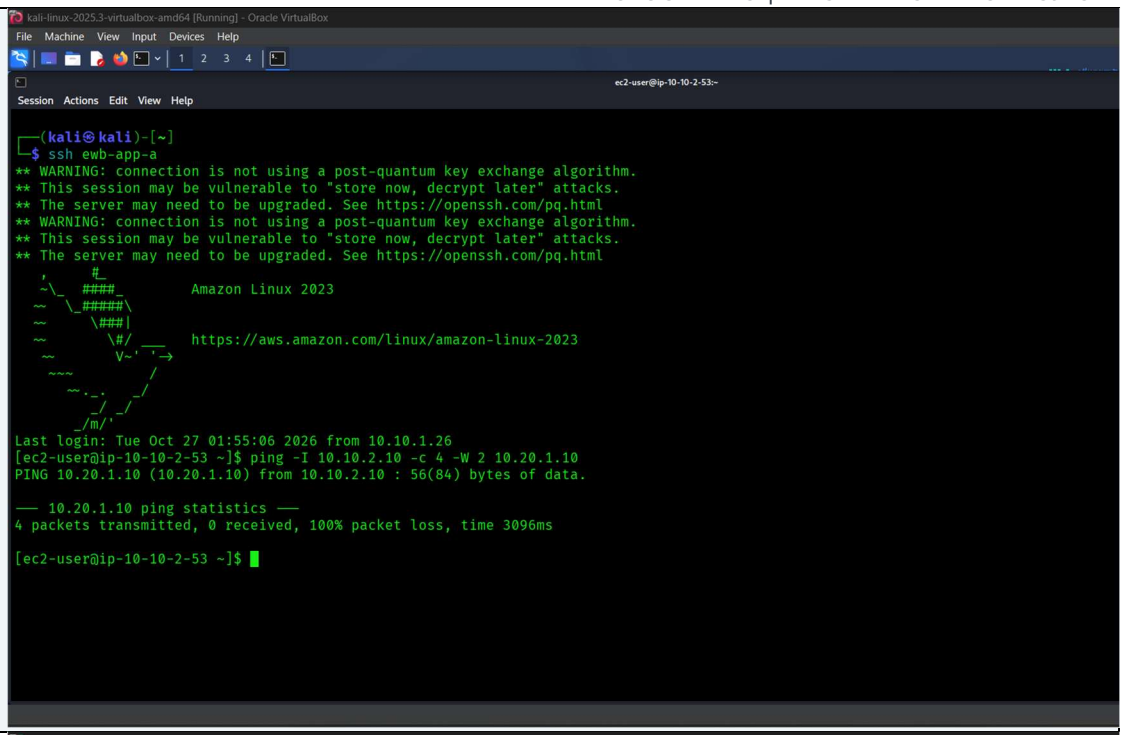
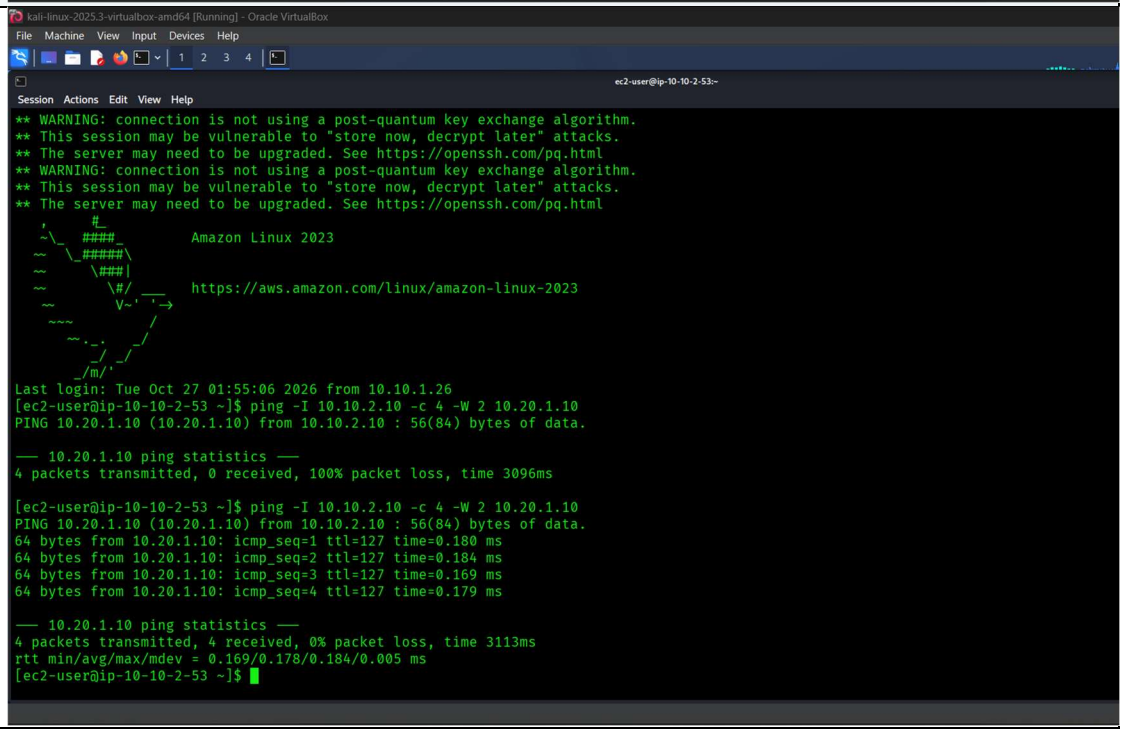

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Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Instances (1/3) Info
Last updated 14 minutes ago
Connect
Instance state
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 ...
Security group name
ewb-bastion
i-0473004da282cca31
Running
t3.micro
3/3 checks passed
ap-south-1a
13.232.111.159
ewb-sg-bastion
i-0473004da282cca31 (ewb-bastion)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
VPC ID
vpc-08092d12e8b63a296 (ewb-vpc-a)
Subnet ID
subnet-02b44cff75db44753 (ewb-a-public)
Availability zone
ap-south-1a
Availability zone ID
aps1-az1
Outpost ID
–
IP addresses Info
Public IPv4 address
13.232.111.159 | open address
Private IPv4 addresses
10.10.1.26
IPv6 addresses
–
Secondary private IPv4 addresses
10.10.1.10
Carrier IP addresses (ephemeral)
–
Hostname and DNS Info
Public DNS
–
Private IP DNS name (IPv4 only)
ip-10-10-1-26.ap-south-1.compute.internal
IPv4-only IP based name: A record
–
Dualstack - IP based name: A and AAAA record
–
IPv6-only - IP based name: AAAA record only
–
Public hostname type
–
Private hostname type
IP name: ip-10-10-1-26.ap-south-1.compute.internal
Use RBN as guest OS hostname
Disabled
Answer RBN DNS hostname IPv4
Disabled
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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07-vpc-peering-active.png	Peering connection status Active	
08-security-groups.png	SSH and ICMP inbound rules	


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Name
Security group ID
Security group name
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Description
sg-09d1174a8b11ca1d9
ewb-sg-app-b
vpc-0cabf8113fbdd7d46
tert
sg-0289d3115775583dc
ewb-sg-bastion
vpc-08092d12e8b63a296
sAS
sg-086fae9dc83bec4b6
ewb-sg-app-a
FGH
sg-09d1174a8b11ca1d9 - ewb-sg-app-b
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Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
sgr-02201dc38efa2ca5d
IPv4
SSH
TCP
22
sgr-07af5791d4f8558af
All ICMP - IPv4
ICMP
All
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Inbound security group rules successfully modified on security group (sg-0289d3115775583dc | ewb-sg-bastion)
Details
Security Groups (1/6) Info
sg-0b4d0a50f9adc7426
default
vpc-08092d12e8b63a296
default VPC security group
sg-0681191bde3ccd1fc
sg-0289d3115775583dc - ewb-sg-bastion
Inbound rules (1)
sgr-08179cb5783b9d616
IPv4
SSH
TCP
22

09-app-a-to-app-b-ping.png	Successful App-A → App-B ping	
10-app-b-to-app-a-ping.png	Successful App-B → App-A ping	

11-ssh-through-bastion-app-a.png	SSH session to App-A through bastion	
12-ssh-through-bastion-app-b.png	SSH session to App-B through bastion	

13-controlled-icmp-failure.png	Controlled ICMP denial, if required	
14-icmp-recovery.png	Connectivity restored after rule recovery	

10. Cleanup and Cost Control

If the lab requires teardown, confirm which resources must be retained before deleting anything. A typical cleanup sequence is:

1. Terminate lab EC2 instances only after required screenshots and logs are saved.
2. Release unused Elastic IP addresses and remove resources that incur ongoing charges.
3. Delete the VPC peering connection after confirming it is no longer needed.
4. Delete dependent network resources in order: instances/ENIs, subnets, route tables, Internet Gateway, then VPCs.

5. Verify the EC2, VPC, and billing consoles to ensure no lab resources remain unintentionally active.

Do not perform cleanup until the required evidence is captured and the lab instructions confirm that teardown is expected.

11. Conclusion

The project design uses VPC peering, reciprocal route-table entries, restricted security-group rules, and bastion-based SSH access to support private communication between two AWS VPCs. Recorded ping results show successful bidirectional connectivity for the intended secondary private IP pair with no packet loss. The final submission should include authentic screenshots for peering status, route tables, security groups, both ping directions, SSH access, and any required failure/recovery or cleanup demonstrations.

12. Final Readiness Checklist

- ☐ Confirm VPC peering status is Active.
- ☐ Verify both workload subnets are associated with the expected route tables.
- ☐ Confirm reciprocal routes point to the peering connection.
- ☐ Verify security-group rules match actual test source IPs.
- ☐ Re-check current bastion public IP and all instance private IPs.
- ☐ Capture actual SSH and ping screenshots.
- ☐ Complete failure/recovery evidence if required by the lab rubric.
- ☐ Update missing IDs, batch details, and any changed addresses.
- ☐ Save this Word document and screenshot evidence in the GitHub project repository.